

# Multinomial Naive Bayes

$$y \in \{0, 1, 2, \dots, k\}$$

In Spam or Non-spam Email :-

$$y \in \{0, 1\}$$

~~$y \in \{0, 1\}$~~

$$\therefore P(y=1|x)$$

$$P(y=0|x)$$

Assumption:

(i) BOV (Ordering doesn't matter)

(ii) Conditional Independence  $\checkmark P(x_i|y) \checkmark$

So, 
$$P(y=1|x) = \overbrace{P(x|y=1)}^{\text{Likelihood}} \cdot P(y=1)$$

$$= \prod_{i=1}^{|V|} P(x_i|y=1) \cdot P(y=1)$$

Here 
$$P(x_i|y=c) = \frac{\text{count}(x_i, y_i=c)}{\sum_{w \in V} \text{count}(w, y_i=c)}$$

*class* (pointing to  $y=c$ )

likelihood (pointing to the fraction)

Ex:

$$P(\text{offer}|y=1) = \frac{\text{count 'offer' in Spam sentences}}{\text{Count all words in Spam sentences}}$$

## Laplace Smoothing

$$V = \{w_1, w_2, \dots, w_n\}$$

$$P(y=1)$$

$$P(y=0)$$

Prior

$$P(w_1|y=1)$$

$$P(w_1|y=0)$$

$$P(w_2|y=1)$$

$$P(w_2|y=0)$$

Conditional probability

$$\vdots$$

$$P(w_n|y=1)$$

$$\vdots$$

$$P(w_n|y=0)$$

$$X\text{-test} = w_1, w_2, \dots, w_n, w_{n+1}$$

$$\therefore P(w_{n+1}|y=c) = \frac{\text{Count } w_{n+1} \text{ in class } c \text{ sentences}}{\text{Count all words in class } c \text{ sentences}}$$

$$= \frac{0}{\text{Something}} = 0$$

Problem

Laplace Smoothing Solution

$\alpha \geq 0$   
(Laplace Smoothing)

Controlled  
Bias Variance  
Trade-off.

$$P(x_i|y_i=c) = \frac{\text{count}(x_i, y_i=c) + \alpha}{\sum_{w \in V} \text{count}(w, y_i=c) + \alpha |V|}$$

Length of  
whole vocab  
(Unique word)



$x_i$  can take count  
 $v$  number of times



# Ex: Multinomial Naive Bayes

|              | Doc ID | Words in document                   | c = China? |
|--------------|--------|-------------------------------------|------------|
| Training Set | 1      | Chinese Beijing Chinese             | yes        |
|              | 2      | Chinese Chinese Shanghai            | yes        |
|              | 3      | Chinese Macao                       | yes        |
|              | 4      | Tokyo Japan Chinese                 | no         |
| Testing      | 5      | Chinese Chinese Chinese Tokyo Japan | ?          |

## Solution

Prior:

$$P(y=1) = \frac{3}{4}$$

$$P(y=0) = \frac{1}{4}$$

$$\begin{aligned} \therefore P(y=1|X) &\propto P(\text{chinese}|y=1)^3 \times P(\text{Tokyo}|y=1) \times P(\text{Japan}|y=1) \times P(y=1) \\ &= \left(\frac{5+1}{8+6}\right)^3 \times \frac{0+1}{8+6} \times \frac{0+1}{8+6} \times \frac{3}{4} \end{aligned}$$

$$= 0.003$$

$$P(y=0|X) \propto P(\text{chinese}|y=0)^3 \times P(\text{Tokyo}|y=0) \times P(\text{Japan}|y=0) \times P(y=0)$$

$$= \left(\frac{1+1}{3+6}\right)^3 \times \frac{1+1}{3+6} \times \frac{1+1}{3+6} \times \frac{1}{4}$$

$$= 0.001$$

## Bernouli Naïve Bayes

1. Bernoulli doesn't talk about the frequency of a feature/word.
2. It is only concerned about whether a word is present or not (1/0).

Likelihood

$$P(x_i | Y_i = c) = \frac{\text{count}(d_i \text{ contains } x_i, Y_i = c) + \alpha}{\text{count}(Y_i = c) + 2\alpha}$$

### Prediction

$$P(Y=1|x) = \prod_{i=1}^{|V|} P(x_i | Y=1)^b \cdot (1 - P(x_i | Y=\text{spam}))^{1-b} \times P(Y=1)$$

posterior probability

Prior

→ Same Dataset (Example):

Prior:  $P(Y=1) = \frac{3}{4}$ ,  $P(Y=0) = \frac{1}{4}$

Here,  $P(\text{Chinese} | \text{yes}) = \frac{2+1}{3+2} = \frac{3}{5}$

$P(\text{Beijing} | \text{yes}) = P(\text{Shanghai} | \text{yes}) = P(\text{Macao} | \text{yes}) = \frac{1+1}{3+2} = \frac{2}{5}$

$P(\text{Japan} | \text{yes}) = \frac{0+1}{3+2} = \frac{1}{5}$ ,  $P(\text{Tokyo} | \text{yes}) = \frac{0+1}{3+2} = \frac{2}{5}$

$$P(Y=1|x) \propto \overset{\text{Chinese}}{\left(\frac{3}{5}\right)} \times \overset{\text{Japan}}{\left(\frac{1}{5}\right)} \times \overset{\text{Tokyo}}{\left(\frac{2}{5}\right)} \times \overset{\text{Macao}}{\left(1 - \frac{2}{5}\right)} \times \overset{\text{Shanghai}}{\left(1 - \frac{2}{5}\right)} \times \overset{\text{Beijing}}{\left(1 - \frac{2}{5}\right)} \times \overset{(\text{yes})}{\frac{3}{4}}$$

$$= 0.005$$

Similarly  $P(Y=0|x) = 0.022$



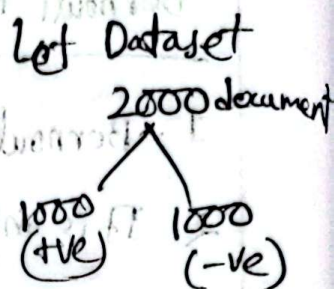
## Bias Variance Tradeoff : Effect of $\alpha$

Consider Bernoulli Naive Bayes,

Case-I: if  $\alpha = 0$

$$P(w_i | y=1) = \frac{2}{1000}$$

only two document  
which contain  $w_i$



If slightly change on Dataset (remove Just 2 document which contain  $w_i$ ),

$$P(w_i | y=1) = \frac{0}{1000} = 0$$

The whole mathematics will be collapse.

So the small change in Dataset leads to a very different model.

So the model has very high variance.

Case-II: if  $\alpha = 10000$

$$P(w_i | y=1) = \frac{2+10000}{1000+20000} \approx \frac{1}{2}$$

Similarly  $P(w_j | y=1) \approx \frac{1}{2}$

So,  $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \dots \times \frac{1}{2} \times P(y=1)$

and  $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \dots \times \frac{1}{2} \times P(y=0)$

Same

So result  
just depends  
on prior.

Which leads  
to Underfitting

When  $\alpha$   
is very  
low,  
Overfitting

Underfitting

Normal Distribution

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

## Gaussian Naive Bayes

→ Used when, features are real Number.

→ Features are Normally Distributed

→ Probability Density function [pdf]

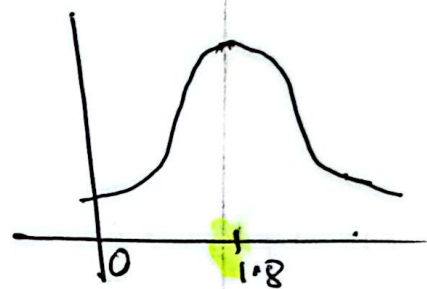
$$P(y=1|X) \Rightarrow \prod_{i=1}^n P(x_i|y=1) \cdot P(y=1)$$

Consider feature (f1),

$$\left. \begin{array}{l} \mu_1 = \text{mean} = 1.8 \\ \sigma_1^2 = \text{variance} = 2 \end{array} \right\} \text{Let}$$

$$P(x_i|y=1) \Rightarrow \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

| f1         | f2 | f3  | f3 | y |
|------------|----|-----|----|---|
|            |    | 168 |    | 0 |
|            |    | 175 |    | 1 |
|            |    | 180 |    | 1 |
|            |    | 158 |    | 0 |
|            |    | ... |    | 0 |
|            |    | 178 |    | 1 |
| height(cm) |    |     |    |   |



Let, X-test = <2.1, 8.2, 1.4, ..., 5.2>

$\uparrow$        $\uparrow$   
 $x_i$      $x_i$   
 for    for  
 f1     f2